

Graph Theory & Networks (AI HL)



Basic graph terminology

- 1 Define the following terms as used in graph theory: vertex, edge, degree of a vertex, and simple graph.
 - 2 A graph has vertices A, B, C, D with edges AB, AC, AD, BC, CD . Find the degree of each vertex, and verify the handshaking lemma (that the sum of degrees equals twice the number of edges).
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Adjacency matrices

- 3 Write the adjacency matrix for the graph with vertices A, B, C and edges AB, BC (undirected, no weights), using the vertex order A, B, C .
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Walks, paths, and connectedness

- 4 Explain the difference between a walk, a path, and a cycle in a graph.
 - 5 Explain what it means for a graph to be connected, and give an example of a disconnected graph with 4 vertices.
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Trees and spanning trees

- 6 State the defining property of a tree (as a type of graph), and find the number of edges in a tree with 7 vertices.
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Applying Kruskal's algorithm for a minimum spanning tree

- 7 A weighted graph has edges with weights: $AB = 4, AC = 2, BC = 3, BD = 5, CD = 6$. Use Kruskal's algorithm (adding the cheapest edge that doesn't create a cycle, in order of increasing weight) to find the minimum spanning tree, and state its total weight.
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Eulerian and Hamiltonian considerations

- 8 State the condition on vertex degrees for a connected graph to have an Eulerian circuit (a closed walk using every edge exactly once), and explain why the classic "Seven Bridges of Königsberg" graph fails this condition.

The Chinese Postman Problem (route inspection)

- 9 A graph representing a delivery route has exactly two vertices of odd degree, B and D , with the shortest path between them having total weight 7. Explain how to find the minimum-weight closed walk that traverses every edge at least once (the route inspection algorithm), and state the total extra weight that must be added.

Applying Dijkstra's algorithm for shortest paths

- 10 A network has direct edges: $A-B = 4$, $A-C = 2$, $B-D = 5$, $C-B = 1$, $C-D = 8$. Use Dijkstra's algorithm reasoning (always extending from the currently cheapest-reached unvisited vertex) to find the shortest path from A to D , and state its total weight.

Prim's algorithm for a minimum spanning tree

- 11 Using the same weighted graph as before (edges $AB = 4$, $AC = 2$, $BC = 3$, $BD = 5$, $CD = 6$), apply Prim's algorithm starting from vertex A (always adding the cheapest edge connecting a vertex already in the tree to one not yet in the tree) to find the minimum spanning tree, and confirm it matches the result from Kruskal's algorithm.

Isomorphic graphs

- 12 Explain what it means for two graphs to be isomorphic, and state two necessary (but not sufficient) conditions that isomorphic graphs must share.

The travelling salesman problem: nearest-neighbor algorithm

- 13 A network has distances $AB = 6$, $AC = 9$, $AD = 4$, $BC = 3$, $BD = 8$, $CD = 5$. Starting at A , use the nearest-neighbor algorithm (always travel to the closest unvisited vertex) to find an upper bound for the shortest route that visits every vertex once and returns to A , and state its total length.

Planar graphs and Euler's formula

- 14 A connected planar graph has 8 vertices and 12 edges. Use Euler's formula for planar graphs, $V - E + F = 2$, to find the number of faces (regions, including the outer, unbounded region).

Weighted adjacency matrices and shortest paths

- 15 A weighted graph has vertices A, B, C with edge weights $AB = 5, AC = 9, BC = 2$ (no direct edge is represented by 0 on the diagonal, and a dash for a non-existent edge). Write the weighted adjacency matrix in the order A, B, C , and use it to state the shortest path from A to C .

Bipartite graphs

- 16 Explain what it means for a graph to be bipartite, and determine whether a graph containing an odd-length cycle can be bipartite.

Complete graphs

- 17 Find the number of edges in the complete graph K_n (in which every pair of vertices is connected by an edge), and hence state the number of edges in K_6 .

In-degree and out-degree in directed graphs

- 18 In a directed graph (digraph), explain the difference between the in-degree and out-degree of a vertex, and find both for vertex B in a digraph with directed edges $A \rightarrow B, B \rightarrow C, D \rightarrow B, B \rightarrow D$.

Chromatic number of a graph

- 19 What is the minimum number of colors needed to color the vertices of a triangle (a cycle of length 3) so that no two adjacent vertices share the same color (i.e. its chromatic number)?
- a. 3
 - b. 2
 - c. 1
 - d. 4

Regular graphs and the handshaking lemma

- 20** A graph is called k -regular if every vertex has degree k . Find the number of edges in a 4-regular graph with 10 vertices, using the handshaking lemma.
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A sufficient condition for a Hamiltonian graph

- 21** State Dirac's theorem, which gives a sufficient condition for a graph to be Hamiltonian (contain a cycle visiting every vertex exactly once), and use it to determine whether a graph with 8 vertices, each of degree at least 4, is guaranteed to be Hamiltonian.